

Embedded Water Quality Analyzer

NOAH BUTLER

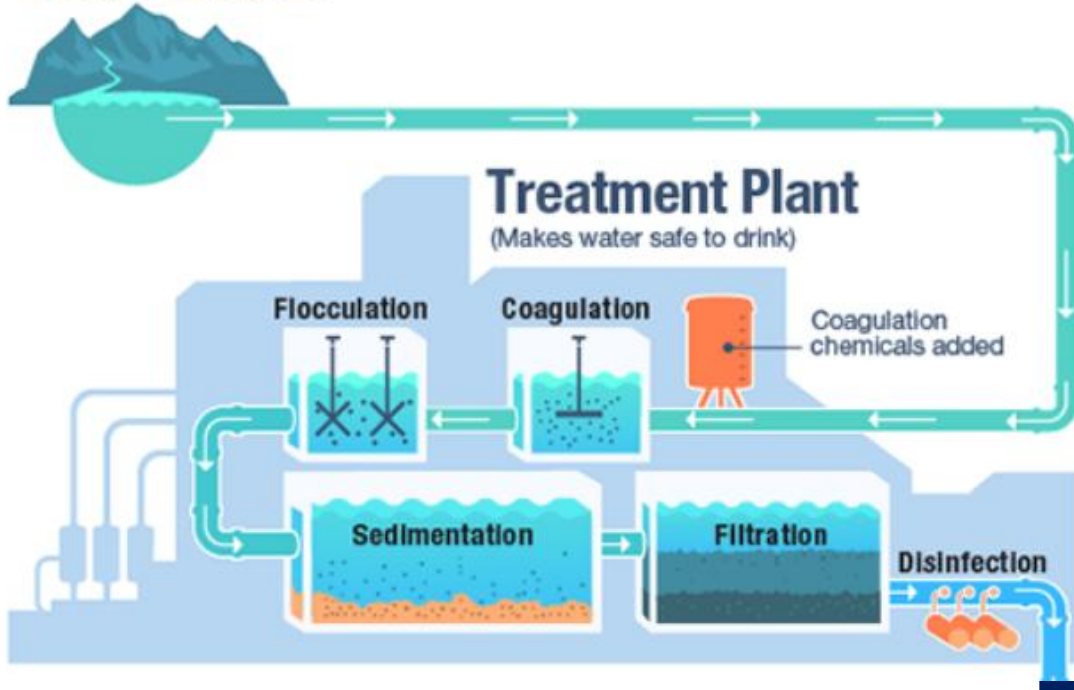
Project Motivation – What's in the Water?

- The rates of cancer particularly among young people in America have been increasing in recent years. Environmental microplastics and the chemicals that they release into the environment have been linked to cancer
- In addition, our health is also influenced by the pH, total dissolved solids (TDS), and other parameters of the water we consume
- As a response, people have a growing interest in what they are consuming and how it may affect their wellbeing
- The Embedded Water Quality Analyzer delivers home water monitoring to its users and identifies potentially hazardous conditions in water that a person could be drinking, showering with, or otherwise consuming

Water is Consumed in Various Ways After Coming From the Water Treatment Facility

1. Water Treatment Plant

Water Source



2. Water in the Home

Drinking & Cooking

Fridge, sink, etc.

Absorption

Shower, Bath, etc.

Indirect Consumption

Vegetable garden hose, etc.

3. Consumption



Various “Forever Chemicals” are Close to the Maximum Limit in a Sample New York Town Water Report

- The highlighted PFAs, or poly-fluoroalkyl substances (better known by the common name “forever chemicals”), are associated with health issues including increased risk of some cancers. There is evidence that PFAs can be leached into the environment by microplastics

PFA Name	Level Detected (ng/l)	Federal MCL ¹ (ng/l)	Federal MCLG ² (ng/l)	Local MCL (ng/l)
Perfluorooctanoic Acid (PFOA)	3.2	4	0	10
Perfluorobutanesulfonic Acid (PFBS)	9.1	--	2000	10
Perfluorooctane Sulfonic (PFOS)	0.87	4	0	10

¹ : MCL = Maximum Contaminant Level ² : MCLG = Maximum Contaminant Level Goal

Sample Water Report Key Points

- No testing for microplastics
 - No treatment for forever chemicals that microplastics leach into the water
 - No testing for pH, or for common dissolved solids like calcium or magnesium
 - Upcoming regulations will put current levels of PFAs into the non-compliant category (some federal MCL goals are as stringent as 0 ng/l)
 - By the time the water is delivered to you, through contact with pipes and delivery systems, it may have changed since it was tested at the water treatment facility
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- **The average health-conscious homeowner may want further testing of their water**

Project Goals

List of Key Project Measurables

The Embedded Water Quality Analyzer should provide its users with the following:

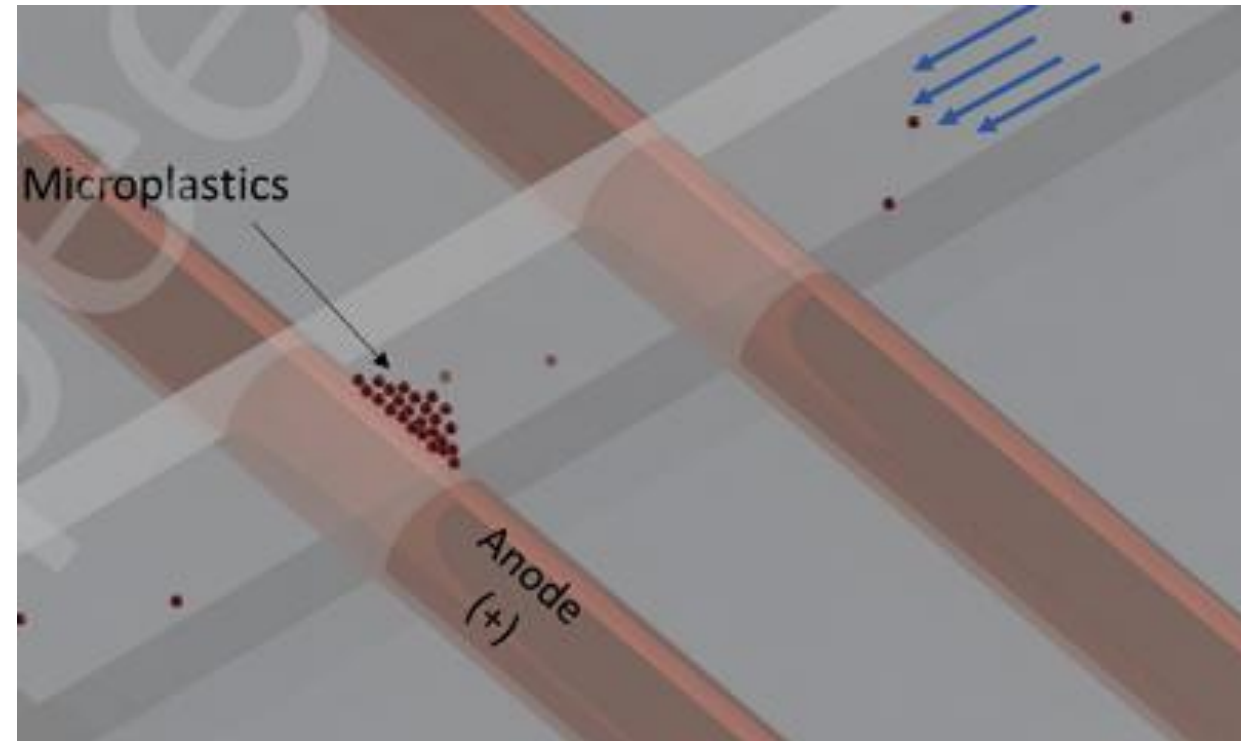
- Accurate ($\pm 5\%$) real-time data of the following water quality markers: temperature, pH, conductivity, and total dissolved solids (TDS)
- Real-time data of turbidity trends
- Real-time data of microplastic concentrations within a $\pm 30\%$ margin of error
- Intuitive ways of visualizing data and trends over time
- System should be based around microcontroller(s) interfacing between hardware modules
- Software should use high-level APIs off which users can add functionality easily

Experiments Show Microplastic Particles Accumulate Around Anodes

Concepts referenced from (Zabihihesari A. et Al., 2022, “Simple Microfluidic Device for...Detection of Microplastics in Water Using Dc Electrical Signal”)

Theory and Findings

- The research paper found that the presence microplastic particles in a solution of de-ionized water reduced electrical resistance between two electrodes
- It concluded that due to the microplastic particles' negative charge, they accumulated around the anode due to the electric field generated between the electrodes



Microplastic Particles Can Differ in Shape, Size, and Electrical Properties Which are Limitations to the Paper's Findings

Main Limitations of the Microplastic Detection Research Paper:

- Research paper uses a different kind of plastic and smaller size of microplastic than used in the testing for this project
- Paper also reveals that microplastic particles can also have a positive charge, however it doesn't quantify the percentage of positively-charge microplastics one could expect in their environment
- Research paper offers no information about differentiating between other micro-particle materials that also may be carrying a charge in the carrier solution

How is a Microplastic Sensor Made?

1. Relationship between resistance, resistivity, and conductivity

- $\rho = RA/l$, where ρ is resistivity, R is resistance, A is cross-sectional area, and l is length
- $\sigma = 1 / \rho$, where σ is conductivity

2. The relationship between microplastic concentration and the conductivity of a buffer solution can be used to estimate the concentration of plastic

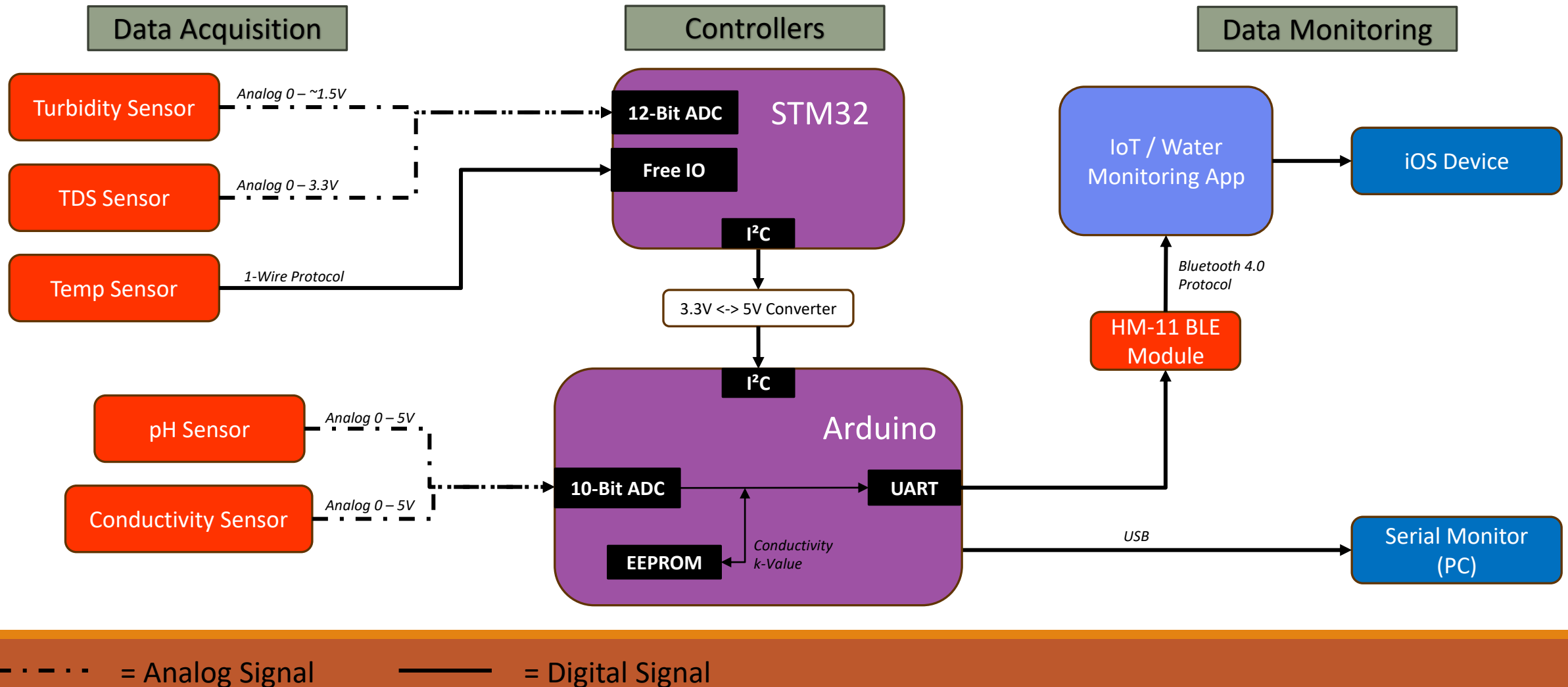
3. Experiments using an embedded conductivity sensor and various solutions of known concentrations of microplastics were used to develop formula for microplastic concentration



**Microplastics (Circled in Red) Shown
Accumulating on Conductivity Sensor Electrode**

Specifications & Overview

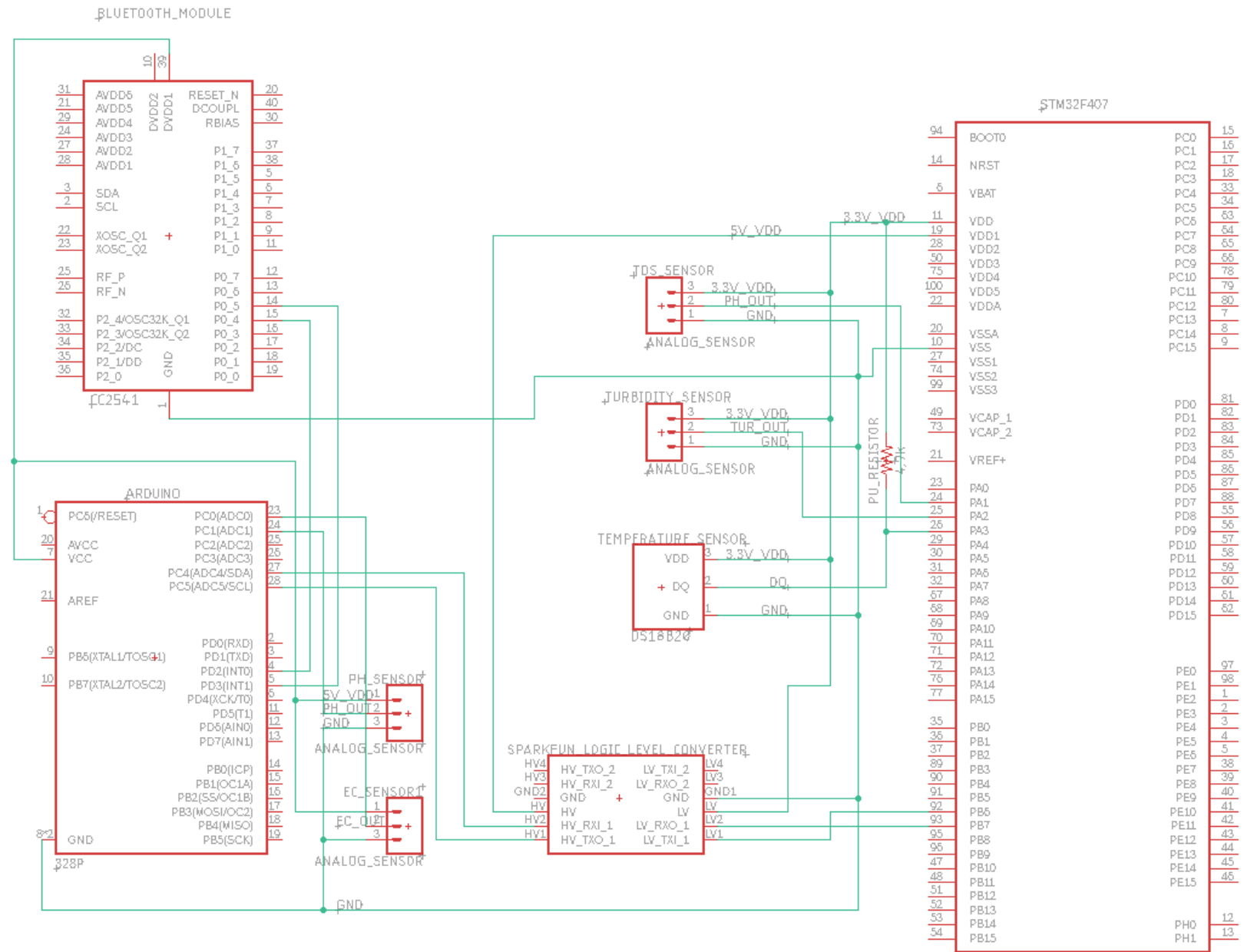
System Functional Overview – 2 Controllers, 5 Sensor Inputs, and Bluetooth Monitoring Output



Bill of Materials

Part Name	Manufacturer	Link	Description	Qty	Cost (No Tax + Shipping included)
STM32F407	STMicroelectronics	https://www.st.com/en/microcontrollers/stm32f407.html	One of the microcontrollers used in project - Arm Cortex-M4 core - Running at 16MHz.	1	\$ 21.17
Arduino Uno R3	Arduino	https://store.arduino.cc/arduino-uno-r3	One of the microcontrollers used in project - ATMEGA328P core - Running at 16MHz.	1	\$ 27.60
Logic Level Converter	SparkFun	https://www.sparkfun.com/products/11222	5V to 3.3V bidirectional logic level converter. Used to interface signals from the 3.3V STM32 to 5V Arduino.	1	\$ 3.50
HM-11 Bluetooth 4.0 BLE Module	DSD Tech	https://www.dsdtech.com/hm-11-bluetooth-4-0-ble-module	MCU peripheral that interfaces communication to other devices via bluetooth 4.0 protocol.	1	\$ 10.99
DS18B20 Temperature Sensor	Gikfun	https://www.gikfun.com/products/18b20-1-wire-digital-temperature-sensor	Embedded temperature sensor. Purchase link includes 4.7kOhm resistor, needed as a pull up on the digital signal line.	Minimum Available Packs of 5	\$ 10.99
Turbidity Sensor	KEYESTUDIO	https://www.keyestudio.com/Keyestudio-KY-021-Turbidity-Sensor-module-p-1122.html	Embedded turbidity sensor. Includes probe and signal processing board.	1	\$ 12.50
Electrical Conductivity Sensor	DFRobot Gravity	https://store.dfrobot.com/ProductDetail.aspx?ID=1122	Embedded conductivity sensor. Includes probe, signal processing board, and buffer solutions of known conductivity for probe calibration.	1	\$ 69.90
Total Dissolved Solids (TDS) Sensor	CQRobot	https://www.cqrobot.com/products/tds-sensor	Embedded total dissolved solids sensor. Includes probe and signal processing board.	1	\$ 11.99
pH Sensor	GAOHOU	https://www.gaohou.com/products/pH-sensor	Embedded total dissolved solids sensor. Includes probe and signal processing board.	1	\$ 31.99
pH Buffer Solution Powder	Nynelly	https://www.nynelly.com/products/pH-buffer-solution-powder	Buffer powder that when mixed with 250mL of distilled water makes solution of known pH.	Pack of 15	\$ 5.99
				Total	\$ 206.62

Hardware Connections Were Prototyped on a Breadboard



Project Electrical Characteristics – Some 5V Analog Outputs, With Significant Power Going to the Bluetooth Module

Component	Input Min. Voltage	Input Max. Voltage	Input Power (mA)	Output Max. Voltage	Output Current (mA)
pH Sensor	5	5	10	5	-
TDS Sensor	3.3	5.5	6	2.3	-
Turbidity Sensor	3.3	5	11	3.3	-
Conductivity Sensor	3.3	5	20	3.4	-
DS18B20 Temperature Sensor	3	5.5	4	N/A (open-drain)	7 (protocol: 1-wire)
HM-11 Bluetooth Module	2	3.6	50	3.3	15
STM32 <-> Arduino I2C Communication	N/A	N/A	0.01	N/A (open-drain)	3 (at 100kbps)

Note: The STM32's ADC cannot accept input voltages greater than 3.3V.

Another board, the Arduino, is needed to process 5V signals.

Why 2 Microcontrollers?

Electrical needs of sensors put strain on the STM32

- STM32 does not have 5V tolerant analog-configured pins
- Total current demand for all components is approaching half of the absolute maximum current draw for the STM32, aiming the circuit design to use less than half of the absolute maximum

Software

- Most of the software used in this project was low-level driver APIs written from scratch for the STM32 that provide application flexibility
- Relied on manufacturer software libraries only for calibrating sensors which rely on the Arduino IDE and for transmitting BLE signals

STM32F4 ADC Driver APIs Were Written for Modularity and Customizability

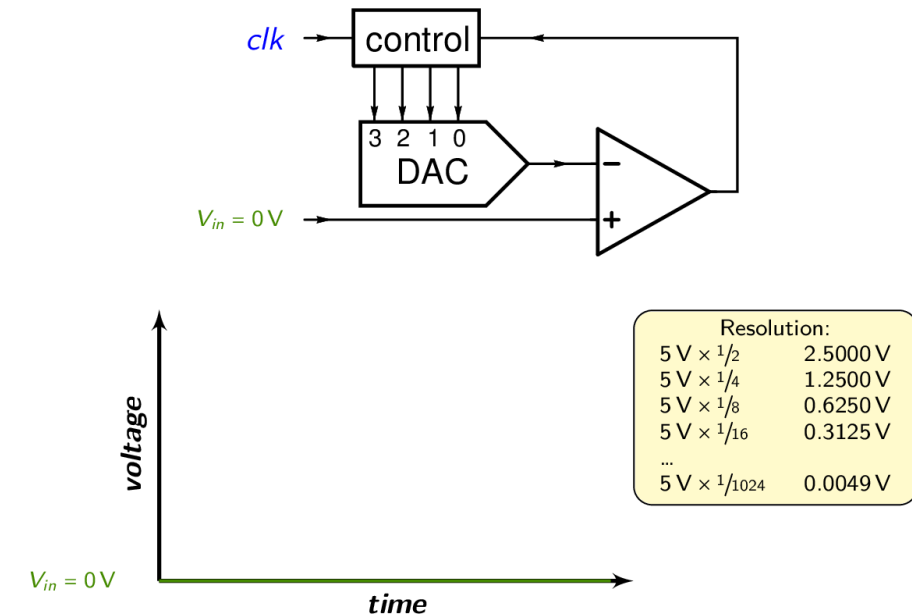
Sensors that output an analog signal: pH, conductivity, turbidity, and total dissolved solids (TDS)

Successive Approximation – example of a 4-bit ADC

STM32F407 ADC APIs and Customizability

- Allows for control over IO pins, clock frequency and timing requirements, interrupt generation, and digital resolution
- Analog watchdog interrupt API allows for immediate application-specific responses to input voltages outside configurable thresholds

In the Embedded Water Quality Analyzer, the STM32 uses 2 input channels in the ADC for turbidity and TDS sensors. The Arduino uses 2 input channels for conductivity and pH



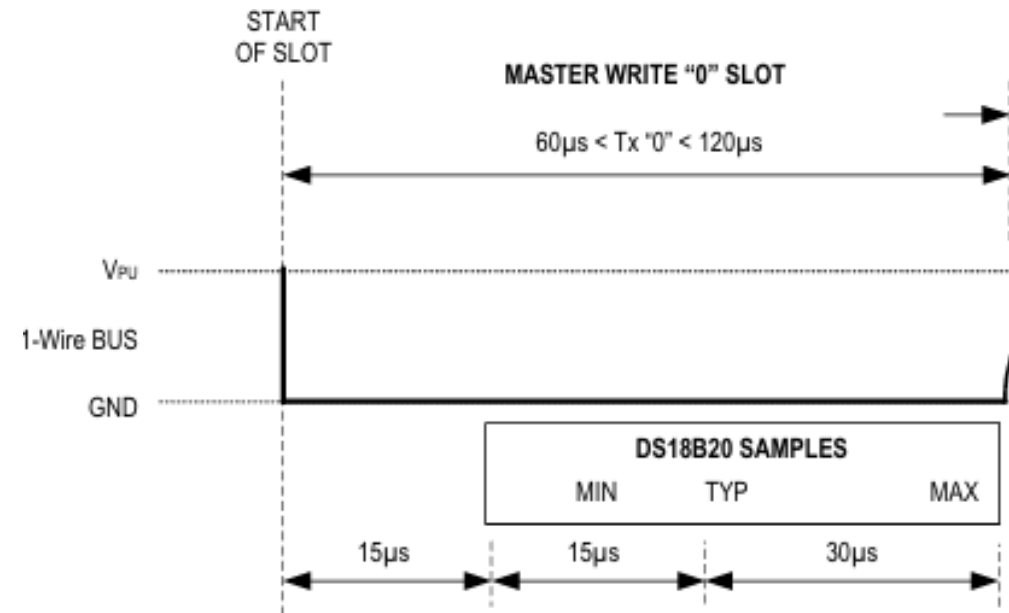
Bit-Banging is Used From the STM32 to Interface With the Temperature Sensor via 1-Wire Communication Protocol

Temperature sensor uses 1-Wire hardware

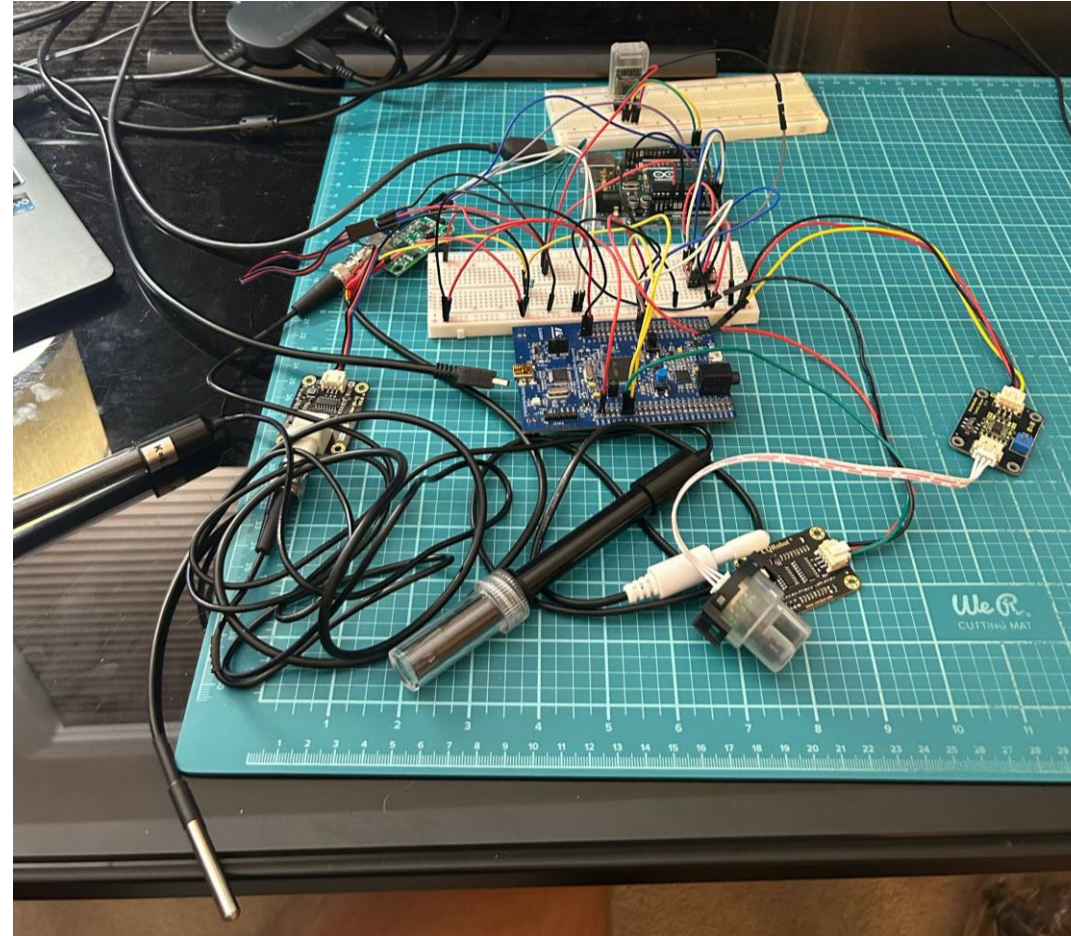
- STM32 interfaces with temperature sensor “scratchpad memory” registers via 1-Wire communication protocol
- Special commands are interpreted by 1-Wire hardware that perform read and write operations over 1 data line

1-Wire makes use of pulses with strict timings

- Reset pulse, presence pulse, write & read 0 or 1 pulses
- After presence pulse from the slave is received, STM32 can send and receive data 1 bit at a time
- Transaction sequence - Initialization, ROM Command, Function Command

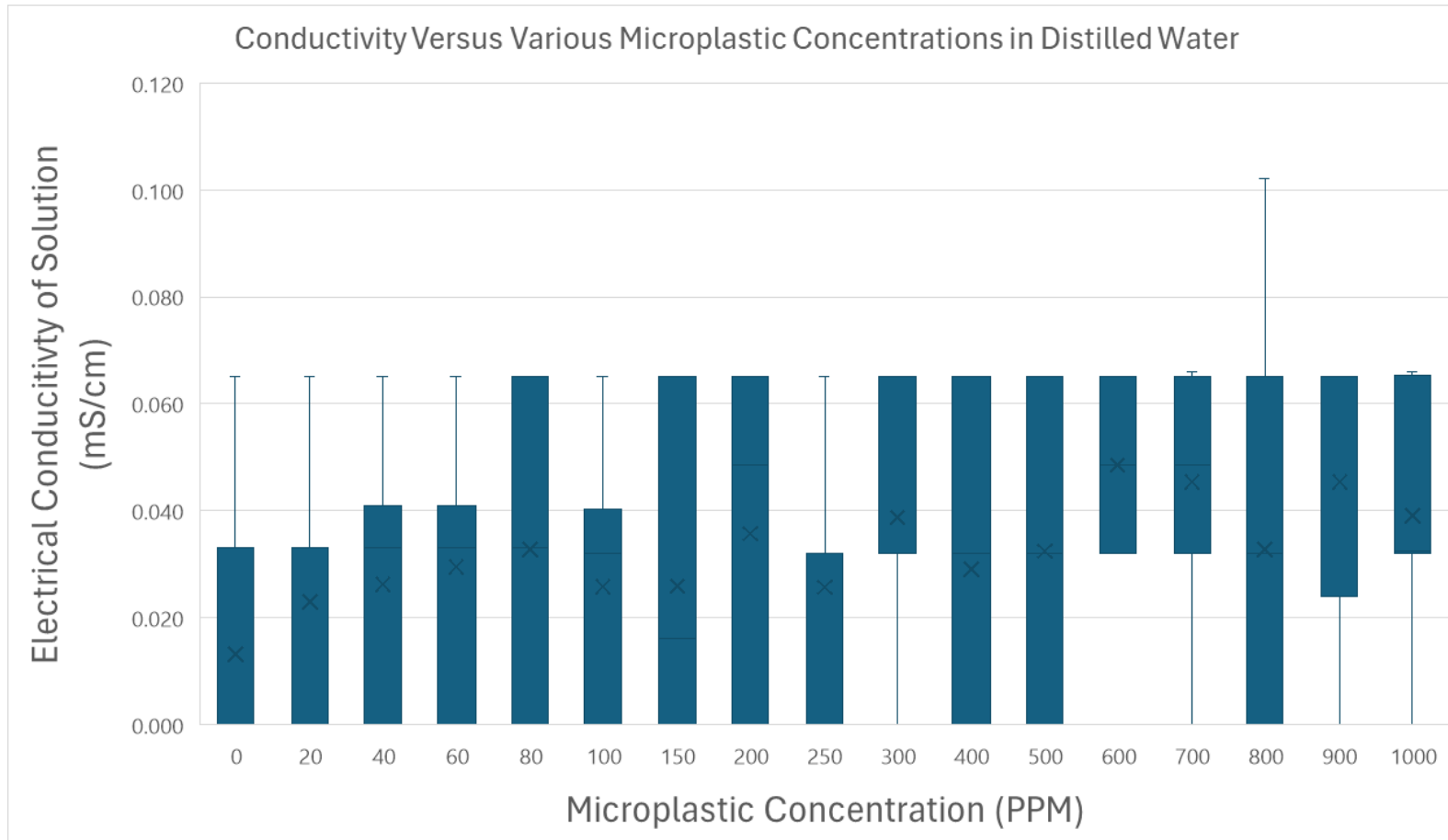


Project Prototype Includes All 5 Sensors, STM32 and Arduino Controllers, and Bluetooth Communication Module



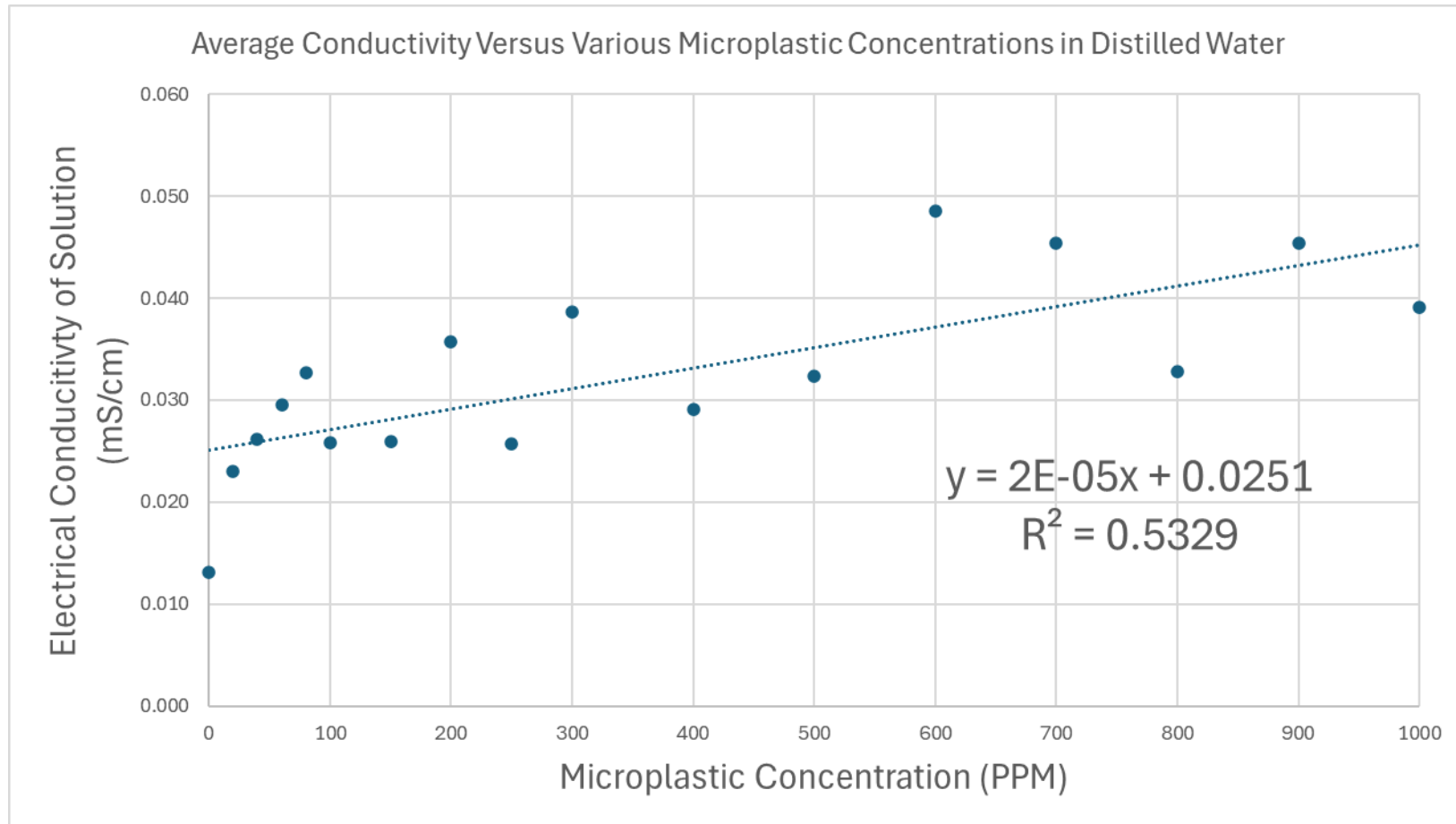
Project Testing

Microplastic Testing – Establishing Relationship Between μ Plastic Concentration and Conductivity



- Created an experimental solution vessel filled with 100mL distilled water
- Microplastic particles of a diameter less than 400 microns were added to the solution, going from 0 to 1000 PPM
- Conductivity measurements (corrected for temperature) were taken at each stage

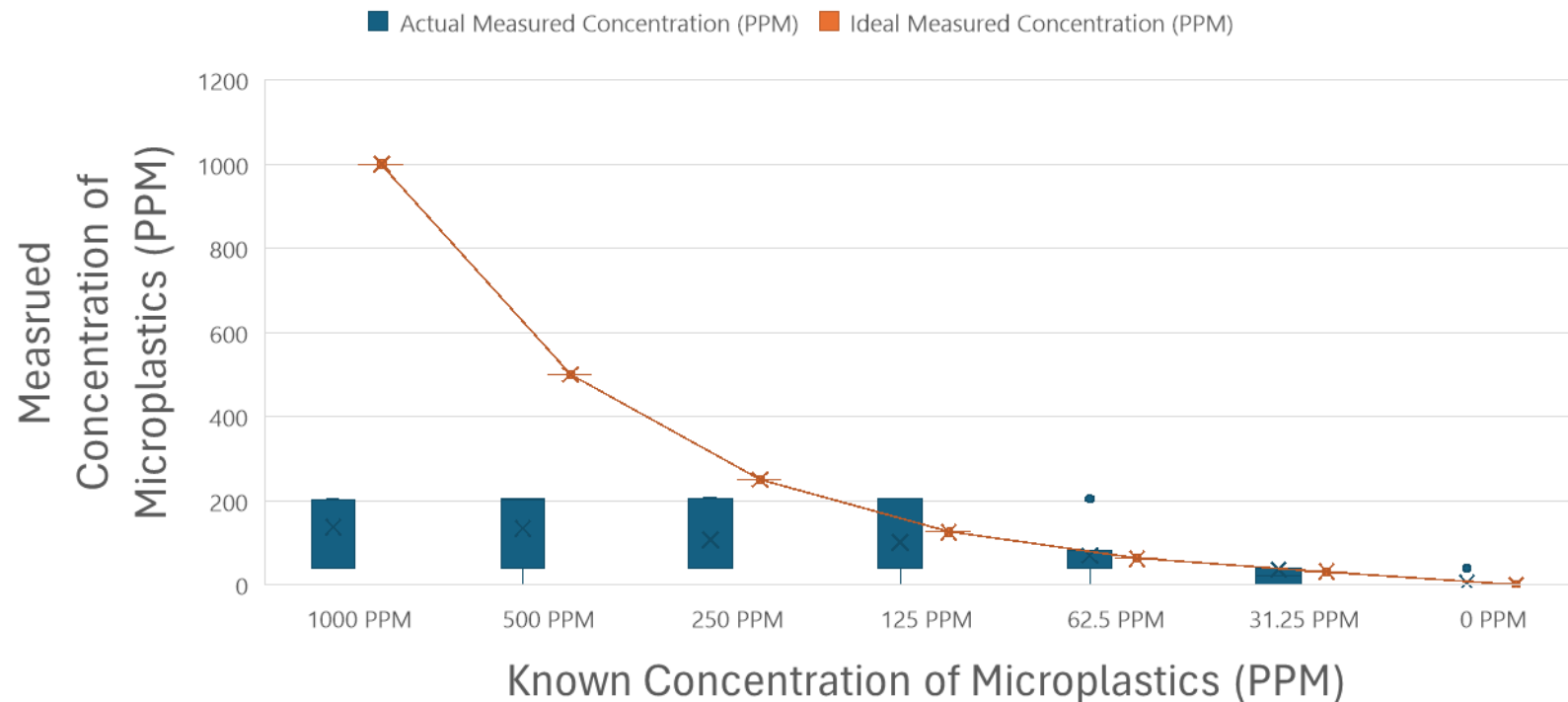
A Linear Line of Best Fit on Average Conductivity for Each Test Condition Shows Decent Correlation



- Microplastic Concentration = $(\text{Conductivity} - 0.025) / 0.0002$
- Returns a zero if a negative value is calculated in microcontroller program implementation
- R-squared value shows decent correlation, with larger concentrations having a “leveling-off” effect

Microplastic Reading Validation Showed Decent Accuracy for Lower Concentrations

Measured Microplastic Concentration Versus True Microplastic Concentration



- Created more solutions of <400µm microplastics from 0-1000ppm for more data points and to test whether the system was accurately reporting the true ppm
- Lower concentrations proved to be easier for the system to accurately report, while system had a harder time with higher concentrations
- Theorized to be due to “saturation” effect of particles on the electrodes

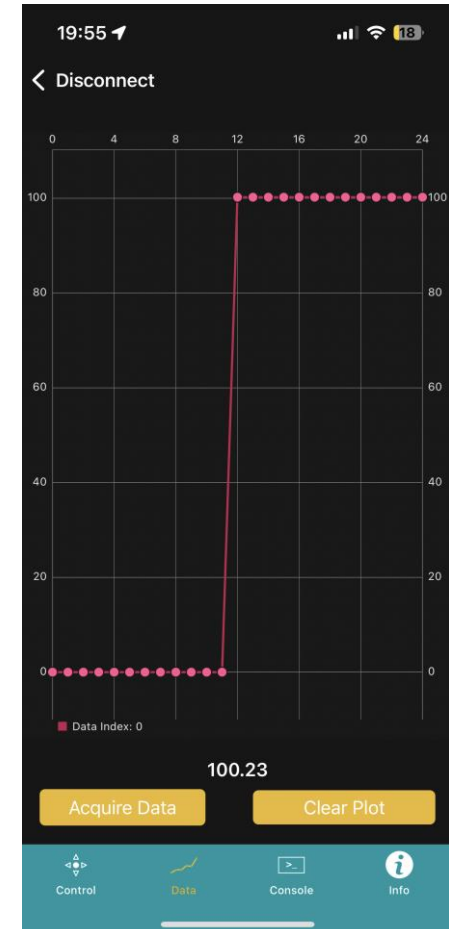
Real-time Data Monitoring via BLE Protocol can Visualize Trends Over Time

Hardware Used – DSDTech HM-11 BLE Module, iPhone 14

Software Used – Maker Portal BLEXAR iOS app, Arduino
SoftwareSerial libraries

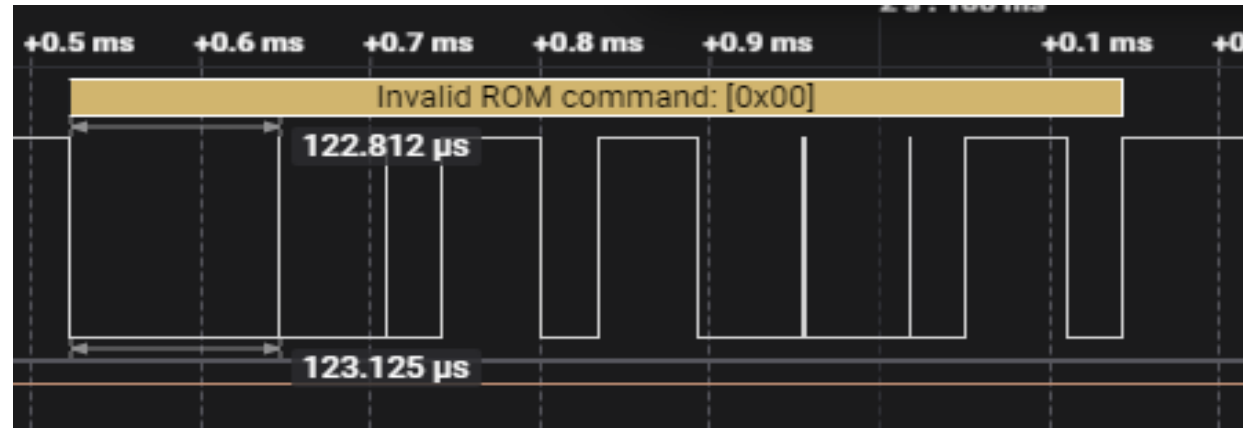
Real-Time Data Trend

- HM-11 BLE Module communicates to master microcontroller over UART, while transmitting and receiving signals over the 2.4GHz ISM band using Bluetooth 4.0 protocol
- Allows for remotely viewing visual trends of the data from up to 30ft away
- This “process dashboard” will allow the user to visually monitor key parameters of their water over time
- If needed, data can be exported as a .csv file to excel



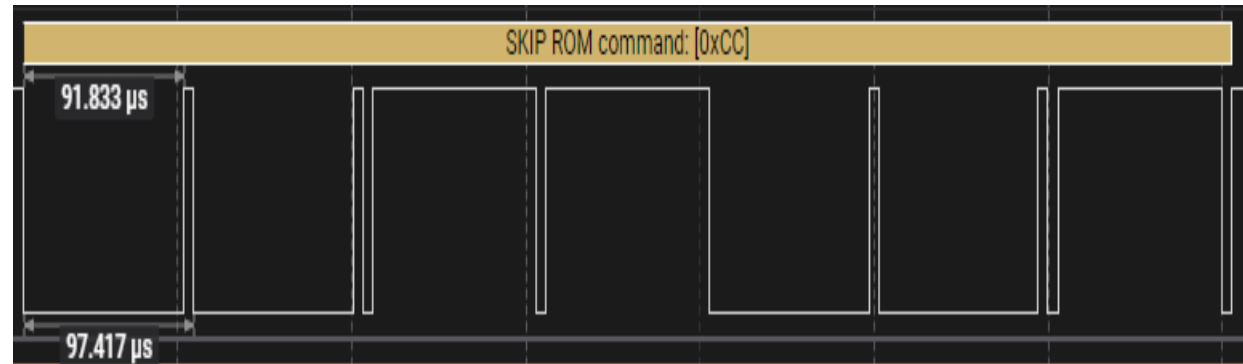
An Example of the Debugging Process Using Signal Analysis on 1-Wire Bus

- **Bug:** In the top figure, there is an attempt to send the ROM code 0xCC. It is invalid because the first bit which is a zero is too long at 122 μ s; it should be between 60 μ s and 120 μ s according to the 1-wire protocol



Firmware Debugged

- **Resolution:** "CEN" (count enable) bit in TIM peripheral register used to control timings was never turned off in between triggers. By resetting the CEN bit while servicing ISR, command is now recognized as the appropriate ROM command



Validation Testing of the Water Quality Analyzer Showed All Project Goals Were Met

1. pH, conductivity, and total dissolved solids (TDS)

- After calibration, both pH, conductivity, and TDS measurements are within $\pm 5\%$ of the true value

2. Temperature

- Within $\pm 0.5^\circ\text{C}$ Accuracy, as specified by manufacturer and proven by thermometer reading

3. Turbidity

- Qualitative data shows correct turbidity trend

4. Microplastic concentration

- Validation of microplastic detection testing showed an average of 13% error in microplastic concentrations less than 125 PPM, with diminishing accuracy in larger concentrations due to “saturation effect”

5. Real-time data logging

- BLE module allows up to four accurate data vs. time trends on the same graph, each with different colors

Project Constraints

Current Design of the Embedded Water Quality Analyzer Limits Its Applications in Some Cases

1. Currently no way to “correct for dissolved ions” in the microplastic calculation
 - This would require the user have a sophisticated filter than they can use to separate a water sample from its microplastics, get a baseline conductivity measurement, and then re-measure the unfiltered water
 - Users should only use the microplastic reading if they are sure of an expected conductivity reading, or if the water is expected to have very little dissolved ions
2. Preciseness of conductivity probe
 - ADC’s resolution only allows for a discrete number of measurements (each step is about .003mS/cm)
3. Inherent wear and tear on conductivity electrodes if in contact with particles – increased equipment degradation
4. Frequent calibrations & need for calibration solutions
 - Both pH and conductivity sensors need regular calibrations with buffer solutions and distilled water

Future Goals

Improvements to the System Include Streamlined Hardware and More Output Functionalities

1. Optimize to a one-controller system

- System only needs one controller that is tolerant of 5V ADC signals, can handle the current demands, and has a flexible set of APIs that allows user to easily add functionality
- STM32F4 and Arduino are overkill for application – a more specialized and lower-cost MCU would take its place in future iterations

2. Improve the resolution of the conductivity reading

- Inherent limitations of low-cost sensor and 10-bit resolution ADC leaves gaps in measurements
- A more precise probe and external higher-resolution ADC would improve microplastic concentration accuracy

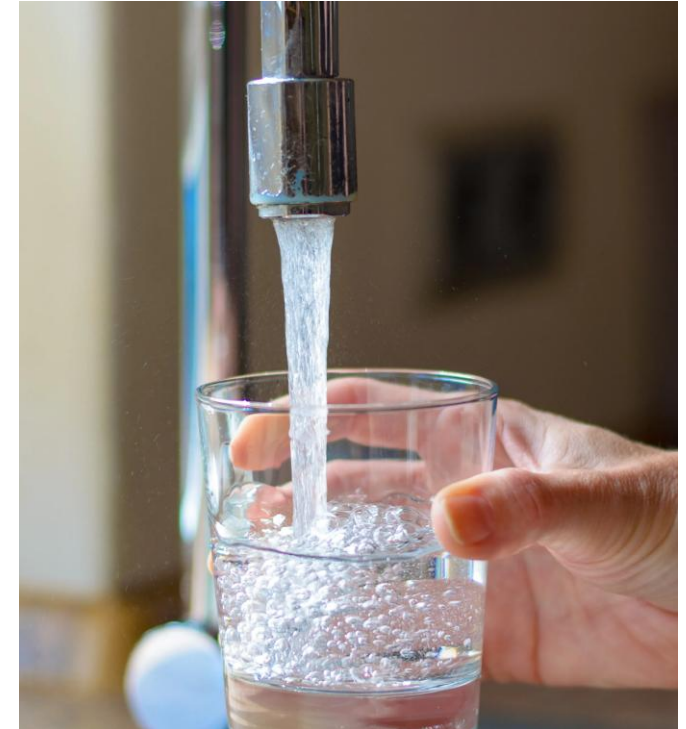
3. Add output functionality

- Alarming – notifications, sounds, etc.; User would be able to configure alarm limits
- Drive motors, servos, etc. on bins for correcting water chemistry tailored to mission critical systems
 - E.g., opening a hatch that adds sodium bicarbonate to balance the pH of an aquaponic fish hatchery

Conclusions

The Embedded Water Quality Analyzer Provides Users Accurate measurements of Their Water and an Intuitive Visualization of the Data

- The Water Quality Analyzer can be used by anyone who is curious about what is really in the water that they consume
- Under ideal conditions, it can provide its users with accurate measurements of the concentration of microplastics in their water
- Remote, real-time monitoring from an iOS device up to 30ft away allows users to easily monitor and identify trends
- Further development towards creating production models will optimize the hardware design and drive down the cost per unit



Sources

- Federal PFA limits:
 - https://www.waterboards.ca.gov/drinking_water/certlic/drinkingwater/pfas.html#:~:text=On%20April%205%2C%202024%2C%20the,which%20MCLs%20will%20be%20adopted
- Microplastic Detection Methodology Research Paper:
 - Zabihisari, Alireza and Khalili, Arezoo and Farshchi-Heydari, Mohammad-Javad and Eilaghi, Armin and Rezai, Pouya, Simple Microfluidic Device for Simultaneous Extraction and Detection of Microplastics in Water Using Dc Electrical Signal. Available at SSRN: <https://ssrn.com/abstract=4215169> or <http://dx.doi.org/10.2139/ssrn.4215169>
- Analog-to-digital converter peripheral visuals:
 - https://en.wikipedia.org/wiki/Successive-approximation_ADC#:~:text=A%20successive%2Dapproximation%20ADC%20is,digital%20output%20for%20each%20conversion